ABHISHEK REDDY MALREDDY

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Machine Learning Engineer experienced in building and deploying production ML pipelines, computer vision systems, LLM applications, and scalable ML APIs. Skilled in Python, PyTorch, SQL, Docker, cloud infrastructure, model evaluation, and end-to-end deployment of AI systems.

EDUCATION

CARNEGIE MELLON UNIVERSITY **GPA: 3.82/4.0** Pittsburgh, PA
Master of Science in Artificial Intelligence Engineering - Materials Science and Engineering Dec 2025
Selected Coursework: Computer Vision, Systems and Tool Chains for AI Engineers, Trustworthy AI, Deep Learning & LLMs.

NATIONAL INSTITUTE OF TECHNOLOGY CALICUT Calicut, India
Bachelor of Technology in Mechanical Engineering May 2022

SKILLS

ML & Computer Vision: PyTorch, Transformers, Scikit-learn, Computer Vision, Semantic Segmentation, Object Tracking, Graph Neural Networks, Reinforcement Learning, Hugging Face
ML Engineering: Model Training, Model Evaluation, Fine-Tuning, Feature Engineering, Hyperparameter Optimization, ML Pipelines, Model Deployment
Languages: Python, SQL, C/C++, TypeScript, JavaScript
Backend & Infrastructure: FastAPI, Docker, AWS, GCP, PostgreSQL, pgvector, Git, Linux
Generative AI: LangChain, LangGraph, RAG, MCP, Whisper, LLM Integration

PROFESSIONAL EXPERIENCE

SPAN Enterprises LLC Fort Mill SC, USA
Junior AI / ML Engineer Dec 2025 – Present

- Designed and deployed an end-to-end ML pipeline processing 9,000+ customer-support calls per month, integrating Whisper transcription, speaker diarization, LLM-based classification, sentiment analysis, and retrieval into a Dockerized production system.
- Built and maintained ML inference workflows using a self-hosted Qwen 8B model to classify topics, detect risky interactions, generate summaries, and support grounded retrieval while keeping sensitive call data within the internal network.
- Created a central CMS in Python and React powering product content and chatbot knowledge for the live TaxBandits.com and Tax990.com applications, with role-based access control and CMS operations exposed as MCP tools.
- Developed a customer-facing chatbot and feedback loop on LangChain and LangGraph with agentic tool calling wired to TaxBandits developer APIs, and fine-tuned models on RunPod to extract structured data from uploaded tax form images.

VIGILANT INC. Remote, USA
Artificial Intelligence Engineering Intern Jun 2025 – Aug 2025

- Designed and deployed a production FastAPI microservice powering GENIE ML Deduplicator for real-time alerts from 10,000+ global sources.
- Engineered a multi-stage ML pipeline using transformer embeddings, DBSCAN, semantic similarity, PostgreSQL, pgvector, and PostGIS.
- Implemented CI/CD workflows for the containerized ML service, including automated testing, Docker builds, deployment validation, and production monitoring.

MACHINE LEARNING & COMPUTER VISION RESEARCH (IIIT-H Affiliated) Hyderabad, India
Research Assistant **Advisor: Dr. Girish Varma** May 2023 – Jun 2024

- **IDD-AW: Robust Semantic Segmentation for Autonomous Driving in Adverse Conditions:**
  - Developed computer-vision models and evaluation methods for semantic segmentation of autonomous-driving scenes across rain, fog, snow, and low-light conditions using a 5,000-image RGB-NIR dataset. Introduced the novel "Safe mIoU" metric to enhance safety evaluation of segmentation models by penalizing critical misclassifications overlooked by traditional mIoU.
  - Built SB3-based reinforcement learning models in SUMO to optimize lane selection and vehicle-to-vehicle communication, reducing emergency vehicle traversal times and surpassing human traffic strategies.
- **Published at the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), Jan 2024.**

ACADEMIC PROJECTS

Machine Learning-Based Characterization of Chemical Ordering in High Entropy Oxides CMU

Research Assistant **Advisor: Dr. Simon Gelin** May 2025 – Dec 2025

- Trained equivariant graph neural network potentials (Allegro, NequIP) on DFT data to predict atomic-scale energies and forces, building ML surrogates that replace quantum-mechanical calculations at a fraction of the compute cost.
- Deployed the trained models as the energy engine for Monte Carlo simulations of HEO-J14, enabling sampling of chemical ordering across temperatures at scales infeasible with direct DFT.

TrackAnything: Multi-modal Promptable Tracking in 3D (CMU) Jan 2025 – Apr 2025

- Built a language-grounded 3D tracking model that maps free-form text prompts (e.g., "walking between parked vehicles") to 3D bounding boxes in driving scenes, enabling text-based scenario mining over the Argoverse 2 dataset.
- Fused text, LiDAR, map, and vehicle pose inputs in a transformer architecture inspired by PromptTrack, aligning language queries with multimodal sensor data.

Real-Time Pedestrian Detection and Safety Alert System (CMU) Jan 2025 – Apr 2025

- Built a real-time pedestrian detection and tracking pipeline using YOLO and DeepSORT for signalized intersections.
- Developed LSTM models for pedestrian intent classification and trajectory prediction to identify potential safety risks.
- Deployed the computer vision pipeline on NVIDIA Jetson Nano using DeepStream SDK for low-latency edge inference.